Tianjian Li

Baltimore, MD | tli104@jhu.edu | Personal Website | Github | Google Scholar

Education

Johns Hopkins University, Ph.D. in Computer Science — Aug 2024 - Present
- Advisor: Daniel Khashabi

Johns Hopkins University, M.S. in Computer Science — Aug 2022 - May 2024
- Advisor(s): Kenton Murray, Philipp Koehn, Daniel Khashabi

New York University, B.A. Joint Major in Mathematics and Computer Science — Aug 2017 - May 2021

Industry Experience

Research Scientist Intern, Meta Fundamental AI Research (FAIR) – New York, NY — June 2026 – Sept 2026
- Manager(s): Weizhe Yuan, Jason Weston.
- Topic: Monitoring reward hacking in auto-research agents. [Blogpost]

Research Scientist Intern, Meta Fundamental AI Research (FAIR) – Bellevue, WA — May 2025 – Dec 2025
- Manager(s): Tianlu Wang, Jack Lanchantin, Jason Weston.
- Topic: Diversity / Exploration in RL post-training.

First-Authored Publications

Self-Compacting Language Model Agents [Paper] [Code]
Tianjian Li, Jingyu Zhang, William Jurayj, Xi Wang, Chuanyang Jin, Mehrdad Farajtabar, Eric Nalisnick, Daniel Khashabi.
NeurIPS 2026.

Jointly Reinforcing Diversity and Quality in Language Model Generations [Paper] [Code]
Tianjian Li, Yiming Zhang, Ping Yu, Swarnadeep Saha, Daniel Khashabi, Jason Weston, Jack Lanchantin, Tianlu Wang.
NeurIPS 2026.

Reasoning over Mathematical Objects: On-Policy Reward Modeling and Test Time Aggregation [Paper]
Pranjal Aggarwal, Marjan Ghazvininejad, Seungone Kim, Ilia Kulikov, Jack Lanchantin, Xian Li, **Tianjian Li***, Bo Liu, Graham Neubig, Anaelia Ovalle, Swarnadeep Saha, Sainbayar Sukhbaatar, Sean Welleck, Jason Weston, Chenxi Whitehouse, Adina Williams, Jing Xu, Ping Yu, Weizhe Yuan, Jingyu Zhang, Wenting Zhao. Preprint 2026.
**Alphabetical order. Led the section on parallel test-time scaling.*

SimpleMix: Frustratingly Simple Mixing of Off- and On-policy Data in Language Model Preference Learning [Paper]
Tianjian Li, Daniel Khashabi.
ICML 2025.

Upsample or Upweight: Balanced Training on Heavily Unbalanced Datasets [Paper][Video]
Tianjian Li, Weiting Tan, Haoran Xu, Kenton Murray, Daniel Khashabi.
NAACL 2025.

Error Norm Truncation: Robust Training in the Presence of Data Noise in Text Generation Models [Paper][Code][Video]
Tianjian Li, Haoran Xu, Philipp Koehn, Daniel Khashabi, Kenton Murray.
ICLR 2024. Spotlight Presentation ($367/7304 \approx 5\%$).

Why Does Zero-shot Cross-lingual Transfer Fail? An Explanation and a Solution [Paper]
Tianjian Li, Kenton Murray. ACL 2023 Findings.

Non-First Authored Publications

ThoughtTrace: Understanding User Thoughts in Real-World LLM Interactions [Paper]

Chuanyang Jin, Binze Li, Haopeng Xie, Cathy Mengying Fang, **Tianjian Li**, Shayne Longpre, Hongxiang Gu, Maximillian Chen, Tianmin Shu.

RLxF Workshop, ICML 2026. Best Paper Award.

Many-Tier Instruction Hierarchy in LLM Agents [Paper]

Jingyu Zhang, **Tianjian Li**, William Jurayj, Hongyuan Zhan, Benjamin Van Durme, Daniel Khashabi.

EMNLP 2026 Findings.

The Flaw of Averages: Quantifying Uniformity of Performance on Benchmarks [Paper]

Arda Uzunoglu, **Tianjian Li**, Daniel Khashabi.

EMNLP 2026.

The Translation Barrier Hypothesis: Multilingual Generation with Large Language Models Suffers from Implicit Translation Failure [Paper]

Niyati Bafna, **Tianjian Li**, Kenton Murray, David R. Mortensen, David Yarowsky, Hale Sirin, Daniel Khashabi.

AACL 2025.

Benchmarking Language Model Creativity: A Case Study on Code Generation [Paper][Code][Video]

Yining Lu, Dixuan Wang, **Tianjian Li**, Dongwei Jiang, Sanjeev Khudanpur, Meng Jiang, Daniel Khashabi.

NAACL 2025.

Verifiable by Design: Aligning Language Models to Quote from Pre-Training Data [Paper][Code][Video]

Jingyu Zhang, Marc Marone, **Tianjian Li**, Benjamin Van Durme, Daniel Khashabi.

NAACL 2025. Oral Presentation.

Technologies

Languages: Python, C++, Java, Shell Script.

Frameworks and Tools: Linux, PyTorch, HuggingFace, Inference Engines (vLLM, SgLang), RL frameworks (verl, slime, prime-rl), Distributed Training (FSDP, Megatron-LM, DeepSpeed), vim, Latex.

Awards

National Olympiad in Informatics Provinces (NOIP) First Prize. 2015

Invited Talks

Beijing Academy of Artificial Intelligence (BAAI) Dec 2025

Johns Hopkins University, CLSP Seminar Spring 2024

Service

Reviewer for —

- NLP Conferences: ACL (2023, 2024, 2025), EMNLP (2023, 2024, 2025), NAACL (2025), EACL (2024, 2025), COLM (2024, 2025, 2026).
- ML Conferences: ICLR (2025, 2026), NeurIPS (2025), ICML (2026).

Organizer: Mid-Atlantic Student Colloquium on Speech, Language and Learning (2024)

Teaching

Head Teaching Assistant, JHU CS 601.471/671 NLP: Self-supervised Models Spring 2025

- Gave lectures on distributed training: [Video 1][Video 2]

Course Assistant, JHU CS 601.471/671 NLP: Self-supervised Models Spring 2024